

Ajay Krishna

Information Technology Undergraduate

Chennai, Tamil Nadu

☎ +917010837356 ✉ prasannaajaykrishna@gmail.com 📠 ajay-krishna-588a1228b 🌐 ajaykrishna00-7

Summary

Final-year Information Technology undergraduate at Madras Institute of Technology, Anna University, with software engineering experience at HSBC and machine learning research experience at IIT Bombay. Experienced in building production monitoring systems, predictive analytics pipelines, and AI-driven applications using Java, Python, and modern ML frameworks.

Education

Madras Institute of Technology, Anna University

Aug 2023 – Jul 2027

B.Tech in Information Technology; CGPA: 8.30

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, AI/ML

Experience

HSBC

Jun 2026 – Jul 2026

Summer Intern

- Engineered a predictive alerting platform for enterprise messaging infrastructure (Solace), delivering it end-to-end (Java backend, React dashboard) with a champion-challenger LightGBM rule engine now in active production use.
- Developed a security remediation copilot that automated fixes for critical open-source vulnerabilities across internal services.

IIT Bombay

Nov 2024 – Sep 2025

Undergraduate Research Fellow

- Researched transformer-based architectures for multi-conversation emotion recognition on multi-party conversational datasets.
- Built and optimized end-to-end data processing, training, and evaluation pipelines, improving classification accuracy through systematic benchmarking and ablation studies.

Projects

Solace Monitoring System | *Java, Python, LightGBM, React, Solace*

- Built a full-stack monitoring platform for Solace messaging infrastructure, currently used in production by the HSBC operations team.
- Designed a champion-challenger predictive rule engine that flags anomalies for proactive alerting and issue detection.
- Built a React dashboard on a Java backend for real-time metrics visualization, enabling faster root-cause analysis during incidents.

CUMTA – AI-Based Smart Traffic System | *YOLOv8, OpenCV, Python*

- Built a real-time AI signal and traffic-pattern analyzer using YOLOv8 and OpenCV on live traffic footage.
- Designed the congestion-detection pipeline as part of a government-backed smart mobility initiative for Chennai.
- Tuned the detection pipeline for real-time inference on live camera feeds to support deployment at city intersections.

Jarvis – AI Desktop Assistant | *Python, Gemini API*

- Developed a modular AI desktop assistant integrating conversational AI, command execution, and persistent memory.
- Built automation, logging, and context-management modules enabling task execution and personalized interactions.
- Designed a plugin-style module architecture, making it easy to add new voice commands and integrations.

Technical Skills

Languages & Tools: Python, Java, C++, SQL

Frameworks & Libraries: TensorFlow, OpenCV, NumPy, Pandas, Transformers

Domains: Machine Learning, Backend Development, Enterprise Software Development

Additional Information

Leadership: AI/ML Lead, Google Developer Group (GDG) on Campus

Research: Co-author of “Residual Affective Word Embedding for Emotion Recognition” (under review)

Certifications: NPTEL – Python for Data Science; NPTEL – Programming in Java; Supervised Machine Learning

Achievements: 1st Place, Prayatna’25 Hackathon (Dept. of Computer Technology); Zonal Champion & Inter-Zonal Runner-up, college chess team (2024)

Languages: Tamil, English, Hindi

Interests: FIDE-rated chess player, astronomy, numismatics, stock market investing